DS 574 **Behavioral Models Notes**
Fall 2026 Prof. Kira Goldner

Behavioral Models

Sometimes, people might come across as irrational, but it can really be explained by varying attitudes toward money or probabilities. Some people prefer certainty to uncertainty. Some people value a dollar less when they already have many dollars. This is the topic we'll cover today.

Which Gamble?

	Gamble A	Gamble B
Original choice	\$10 for sure	\$20 with probability 1/2; \$0 otherwise
Add \$990	\$1000 for sure	\$1010 with probability 1/2; \$990 otherwise

Which gamble do you prefer in each row? What if the \$20 payment were instead \$50 or \$200 (so the corresponding payment in the second row becomes \$1040 or \$1190)? Does adding \$990 to every outcome change your preference?

With utility linear in money, the person is indifferent when the gamble pays \$20 or \$0 and prefers the gamble when the possible gain rises to \$50 or \$200. Adding the same sure payment leaves these comparisons unchanged. Other preferences can reflect nonlinear utility of wealth. The useful comparison is how choices change with the possible gain and with the guaranteed payment.

Expected utility. Let w be the money we start with, and let m describe how much we value the amount we end up with. A gamble adds x dollars with probability q and y dollars with probability $1 - q$. Its expected utility is

$$q\, m(w + x) + (1 - q)\, m(w + y).$$

In the \$20-or-\$0 gamble, $x = 20$, $y = 0$, and $q = 1 - q = 1/2$; $w + x$ and $w + y$ are the resulting amounts of money. For example, $m(z) = \sqrt{z}$ for $z \geq 0$ has decreasing marginal value for money.

Small and Large Stakes

Consider two fair coin flips. Declining either gamble leaves wealth unchanged.

	Heads	Tails
Small gamble	Gain \$11	Lose \$10
Large gamble	Gain \$200	Lose \$100

Would you take each gamble? What if the large gamble offered a gain of \$1000 or \$10,000, still with a possible loss of \$100? Could a large enough gain make it attractive?

A calibration puzzle. Rabin's calibration result [**?**] says: if an increasing, concave utility function over total wealth rejects the small gamble at *every initial wealth level*, it also rejects the large gamble for every finite possible gain. The assumption concerns choices across wealth levels, not one rejection at one wealth level. This challenges explaining persistent small-stakes risk aversion solely by diminishing marginal utility of wealth.

The Allais Paradox

Choose between A and B, then between C and D.

- **First choice.** A: \$1 million for sure.
  B: a 10% chance of \$5 million, an 89% chance of \$1 million, and a 1% chance of nothing.
- **Second choice.** C: an 11% chance of \$1 million; otherwise nothing.
  D: a 10% chance of \$5 million; otherwise nothing.

Preferring A to B and D to C cannot both be explained by expected-utility maximization with the same utilities for the monetary outcomes. This conclusion concerns that pair of strict preferences; either choice alone is consistent with expected utility.

These departures from expected utility motivate cumulative prospect theory [**?**]. We will focus on loss aversion, one part of that broader theory.

Loss aversion alone does not resolve the Allais pattern: cumulative prospect theory also changes how probabilities enter the evaluation.

Loss Aversion

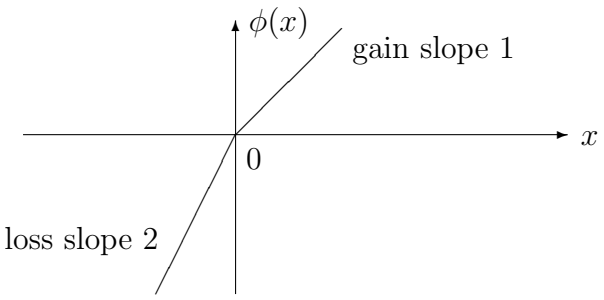
For now, assume $m(x) = x$.

The small-stakes puzzle suggests another possibility: the agent evaluates *changes* from a reference point. A *reference point* is the position used to decide what counts as a gain or loss; here it is the status quo. *Loss aversion* means that losses weigh more heavily than gains of the same size.

Let x be a change from the reference point, measured in dollars. We use a simple value function $\phi(x)$: a gain has weight 1, and a loss has weight $\lambda \geq 1$. The agent averages $\phi(x)$ using the actual outcome probabilities and compares with $\phi(0) = 0$ for no change.

$$\phi(x) = \begin{cases} x & \text{if } x \geq 0, \\ \lambda x & \text{if } x < 0. \end{cases}$$

For $\lambda > 1$, the loss side is steeper. Setting $\lambda = 1$ gives the monetary-change benchmark. This model isolates loss aversion; it does not include diminishing sensitivity or probability weighting.



How much gain compensates for a loss? Set $\lambda = 2$. Return to the fair coin flips: gain \$11 or lose \$10; gain \$Y or lose \$100. Does the agent accept the first? What is the smallest Y for which the second is at least as good as declining? Their expected value for the small gamble is

$$\tfrac{1}{2}\phi(11) + \tfrac{1}{2}\phi(-10) = \tfrac{1}{2}(11) - \tfrac{1}{2}(20) = -4.5 < 0.$$

For the large gamble, it is $Y/2 - 100$, so the agent accepts weakly when $Y \geq 200$. More generally, for a fair gamble gaining $g > 0$ or losing $\ell > 0$, the acceptance condition is

$$\tfrac{1}{2}g - \tfrac{1}{2}\lambda\ell \geq 0 \quad \Longleftrightarrow \quad g \geq \lambda\ell.$$

The reference point is initial wealth for each decision. Rejecting the small gamble can persist as wealth changes without requiring the curvature of one fixed utility function over total wealth to explain every decision. A loss remains a loss even if the final wealth level is large.

The Endowment Effect

Coffee mugs are randomly allocated to 22 of 44 students. Each student wants at most one mug. Suppose values are distinct, independent of initial ownership, and unchanged by transfers of money, and costless trade puts the mugs with the 22 students who value them most. How many mugs should change hands, in expectation?

Each of the 22 highest-valuing students has probability 1/2 of initially receiving a mug. Thus the expected number of these students initially without one is $22/2 = 11$. These are the mugs that must change hands in a direct reallocation to the highest-valuing students.

In the four mug markets of the 44-student study, the observed numbers of trades were 4, 1, 2, and 2, against the benchmark of 11. Comparable induced-value token markets produced 12, 11, and 10 trades against the same benchmark [**?**]. This comparison makes transaction costs common to both types of markets an insufficient explanation of the difference. In the classic mug study, selling prices exceeded buying prices and substantially fewer mugs traded than the ownership-independent benchmark. The *endowment effect* is the tendency to require more compensation to give up an object one owns than one would pay to acquire the same object.

Buying versus selling. Consider the same person with the same underlying mug value $v = 10$ and loss multiplier $\lambda = 2$. Measure a change of one mug as v dollars of value. Evaluate the change in mugs and the change in money *separately* using ϕ, then add. The reference point is current holdings; making no trade has value 0. Assume the person can afford all prices considered.

Buying. The person starts without a mug and can pay a price $p \geq 0$ to acquire one. Identify the gain and the loss. What is the largest price they will pay?

Buying gives a mug gain of v and a money loss of p. The value of trading relative to doing nothing is

$$\phi(v) + \phi(-p) = v - \lambda p.$$

The maximum willingness to pay is $v/\lambda = 5$.

Selling. The person instead starts owning the mug and can receive p to give it up. Identify the gain and the loss. What is the smallest price they will accept?

Selling gives a money gain of p and a mug loss of v. The value of trading is

$$\phi(p) + \phi(-v) = p - \lambda v.$$

The minimum willingness to accept is $\lambda v = 20$. At a price of \$10, this person would neither buy a mug they lack nor sell one they own. Their underlying v and λ are unchanged; ownership changes the reference point.

At $\lambda = 1$, both thresholds equal v. With $\lambda > 1$, the two-dimensional comparison matters: applying ϕ only to the net change $v - p$ would give a buying threshold of v and would miss this mechanism. The numbers here illustrate the model; they are not estimates from the mug study.

Present Bias

Suppose you have not had anything to drink for four hours and have just eaten salty potato chips. Which would you choose in each row?

Choice A	Choice B
1 sip now	2 sips in 5 minutes
1 sip in 20 minutes	2 sips in 25 minutes

Both choices involve waiting five extra minutes for the second sip. Does delaying both options change your preference?

The first option can be particularly attractive when it is available immediately. Present bias models this extra weight on the present [**?**]. For costly tasks, the corresponding pattern is a plan to do the work later that is revised when the work becomes immediate.

Sending a Package

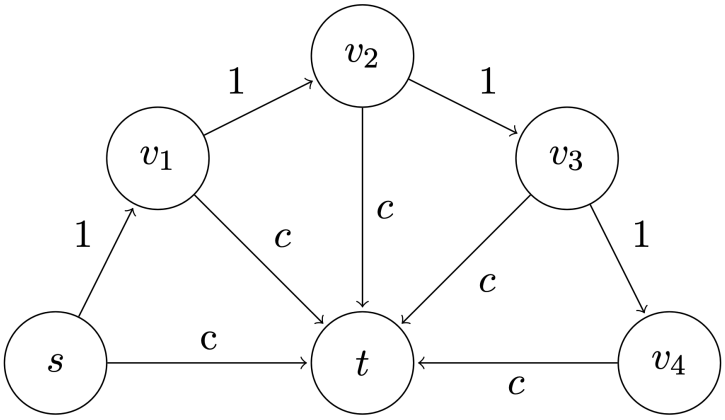
George Akerlof described postponing mailing a package to Joseph Stiglitz. Sending the package has a fixed effort cost c, and each day of delay costs 1 in lost use. Sending after k days therefore has actual total cost $c+k$. An agent minimizing actual cost sends immediately.

A present-biased agent gives an immediate cost extra weight $b \geq 1$, while leaving future costs unchanged. They compare sending today with waiting one day and sending tomorrow. Only today's cost is multiplied by b.

Today versus tomorrow. What are the two perceived costs? When does the agent prefer waiting? Use $b = 2$ and $c = 3$ to compare the choices today, then make the comparison again tomorrow.

Sending today has perceived cost bc. Waiting one day and then sending has perceived cost $b+c$: today's delay costs b, and tomorrow's mailing costs c. Waiting is strictly preferred when $bc > b + c$.

For $b = 2$ and $c = 3$, sending today costs 6 in the comparison, while waiting costs $2 + 3 = 5$. Tomorrow the same comparison recurs. Each day's plan is to send the next day; the agent does not carry out that plan when it becomes current.



Here s is today, v_j is day j, and t means the package has been mailed. Each edge of cost 1 delays mailing by one day; each edge of cost c sends it. Suppose the package must be mailed by day 4.

The agent waits until v_4 and then mails the package, since there is no later mailing day. Actual cost is $4 + 3 = 7$, compared with 3 for sending at s. The factor b changes perceived costs, not actual effort [**?** **?**].

The graph rule. More generally, vertices describe progress toward a task and edges describe possible next actions. The graph is directed and acyclic, with start s, goal t, and nonnegative edge costs. Write $c(v, u)$ for the cost of taking an edge from v to u, and $d(u, t)$ for the ordinary cheapest remaining cost from u to the goal. At v, the perceived cost of taking that edge and then finishing is

$$b\,c(v, u) + d(u, t).$$

The agent chooses an outgoing edge minimizing this expression, then recomputes at the next vertex. They plan as if future choices minimize actual remaining cost; they do not anticipate their future re-planning. This is a *naive* present-biased agent. Fix a tie-breaking rule before any choices, and consider only vertices with a path to t.

For $b = 1$, the expression is $c(v, u) + d(u, t)$, so choosing a minimizing edge follows an ordinary cheapest path. For $b > 1$, the planned continuation can change when its first edge becomes immediate. This distinction between a planned route and a realized route is essential in the deadline example.

Deadlines and Commitment

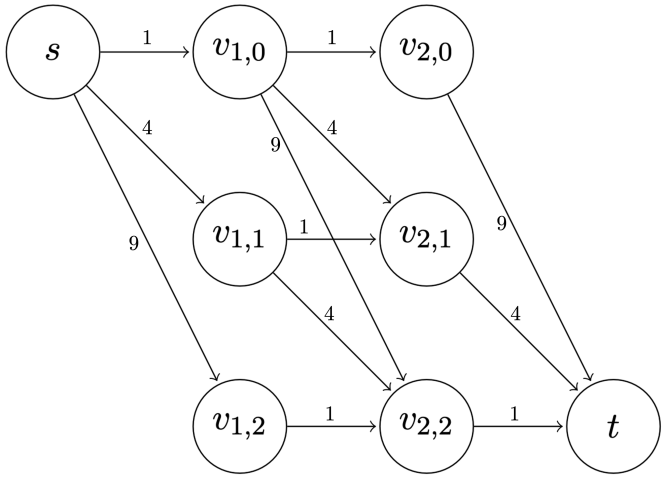
Suppose students can choose binding submission deadlines for their assignments, up to the end of the course. A student who can always follow through on their plans can keep all options available by choosing the latest deadlines. What would you choose? Why?

An earlier deadline can serve as a commitment: it removes a future opportunity to postpone. Whether it helps depends on the choices the student will actually make. We can check this in the task graph below [**?**].

Two Assignments in Three Weeks

There are two assignments and three weeks. The effort cost in a week is 4 for completing one assignment, 9 for completing both, and 1 for completing none. The cost 1 applies even if both assignments are already done. In the graph, $v_{i,j}$ means that i weeks have passed and j assignments are complete; $s = v_{0,0}$ and $t = v_{3,2}$.

The student has $b = 2$ and receives a reward $R = 17$ only on reaching t. At each vertex, compare the least perceived remaining cost with R: continue if it is at most R, and otherwise abandon the course, receiving no reward. Past costs are already incurred and do not enter this comparison.



Without a deadline. At s, compare doing zero, one, or two assignments this week. What does the student plan to do later? Then recompute at each vertex they actually reach. Do they finish? At s, the three perceived costs are

$$\begin{aligned} \text{Do none:} \quad & 2(1) + 4 + 4 = 10, \\ \text{Do one:} \quad & 2(4) + 1 + 4 = 13, \\ \text{Do both:} \quad & 2(9) + 1 + 1 = 20. \end{aligned}$$

The student starts, since $10 \leq 17$, and takes $s \to v_{1,0}$. Their plan at s was to do one assignment in each of weeks 2 and 3.

At $v_{1,0}$, however, the comparison becomes

$$\begin{aligned} \text{Do none:} \quad & 2(1) + 9 = 11, \\ \text{Do one:} \quad & 2(4) + 4 = 12, \\ \text{Do both:} \quad & 2(9) + 1 = 19. \end{aligned}$$

They again postpone, taking $v_{1,0} \to v_{2,0}$. Now both assignments must be done at once, with perceived cost $2(9) = 18 > 17$. The student abandons the course. The path is $s \to v_{1,0} \to v_{2,0}$, followed by stopping; actual incurred cost is $1 + 1 = 2$, with no reward.

Can a deadline help? What deadline would you impose? Which vertex or edges would it remove? Recompute the student's choices in the retained graph, including the cheapest future costs. Does the student finish, and what actual total cost do they incur? Require at least one assignment by the end of week 2. This deletes $v_{2,0}$ and its incident edges. At s, the cheapest planned completion after waiting is still one assignment in each of weeks 2 and 3, so the three perceived costs remain $10, 13, 20$. The student still waits for one week.

At $v_{1,0}$, another week with no assignment is unavailable. Doing one assignment costs $2(4) + 4 = 12$, whereas doing both costs $2(9) + 1 = 19$. The student chooses one, reaches $v_{2,1}$, and then completes the last assignment because $2(4) = 8 \leq 17$. Thus the route is

$$s \to v_{1,0} \to v_{2,1} \to t, \qquad \text{actual cost } 1 + 4 + 4 = 9.$$

The deadline changes feasibility, not the student's bias, effort costs, or reward. Measured by actual reward minus actual effort, the outcome changes from -2 to $17 - 9 = 8$.

Why can fewer options help? An agent who minimizes actual total cost could already choose the deadline-compatible path. Removing options cannot improve the minimum cost available to that agent. The present-biased student instead chooses again at each vertex; removing the option to postpone a second time changes their realized path.

This example proves that a deadline *can* improve completion. It does not show that every deadline helps. A deadline that removes every route to the goal prevents completion entirely. The design question is which restrictions preserve a feasible route and induce the agent to follow through.

Ownership changes which outcomes count as gains and losses. Time changes which cost is immediate. In both examples, predicting behavior requires the relevant reference point or decision time, and the design must account for it.

Acknowledgements

This lecture combines material from the original behavioral economics lectures. It was developed using Shengwu Li's 2019 tutorial on Behavioral Economics and Imperfect Rationality at ACM EC and Sigal Oren's lecture notes. The package and assignment graphs are retained from those notes; the graph model is due to Kleinberg and Oren.